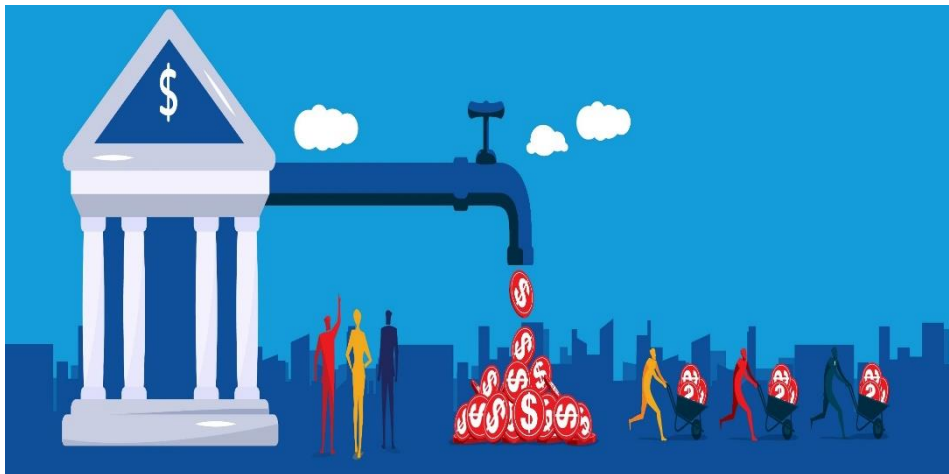


Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

Prof. Francesco Bripi

Central Banks and Monetary Policy (Chapter 10)



Outline of today's lecture

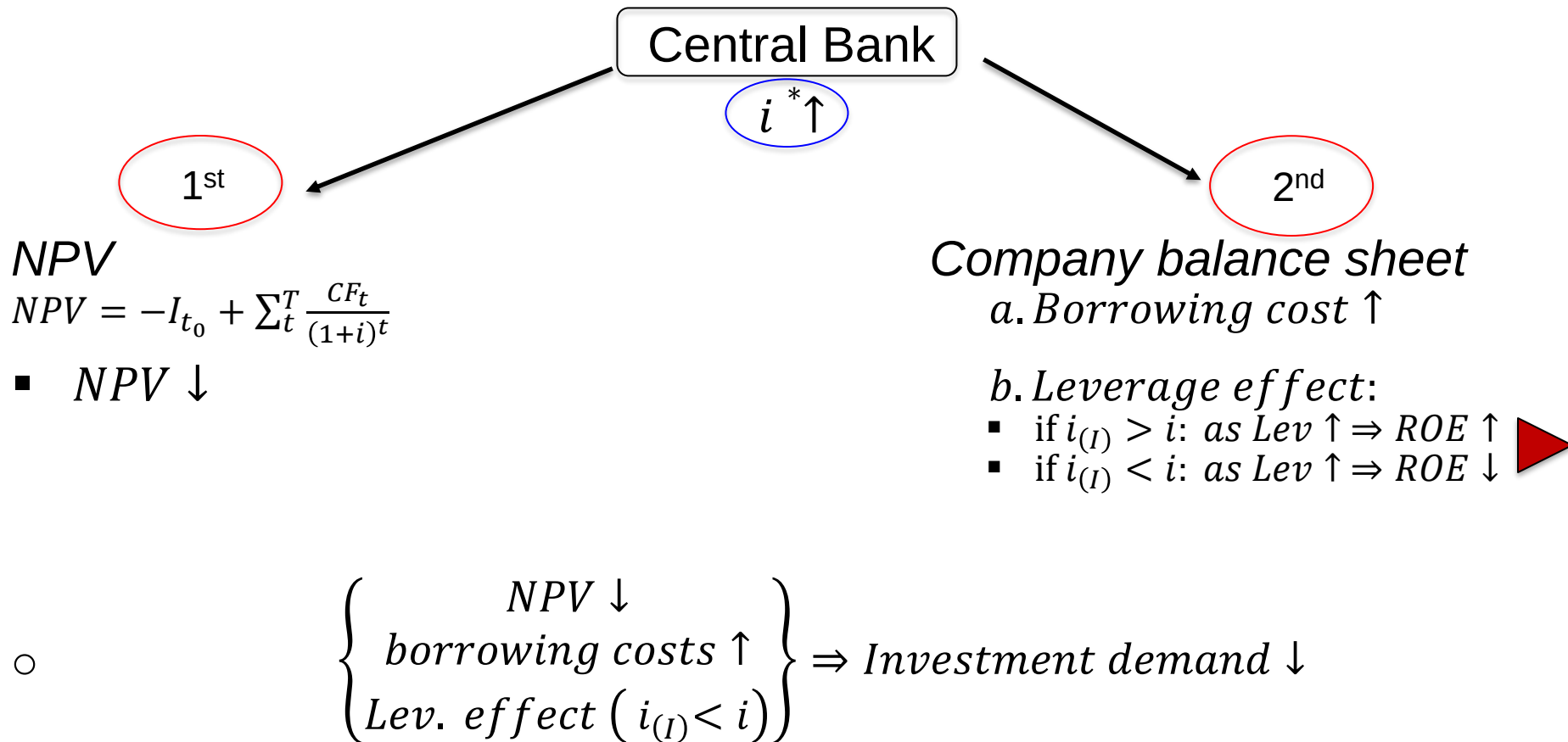
1. Central banks: what they do
 2. Central banks: Balance sheet
 - assets
 - liabilities
 3. Monetary policy instruments
 - normal times
 - in crisis (Unconventional monetary policy)
- ✓ Mostly look at Fed
- ✓ Quick look at ECB

What do Central Banks do

- ✓ Central banks are the “government” authorities in charge of Monetary Policy
- ✓ You studied them in macro already: they have to do with
 - interest rates
 - money supply
 - inflation
- ✓ But they also have effects on **banks**
- ✓ Actually, most of what central banks do goes *through* the (commercial) banking system
- ✓ Important to understand how this works

Monetary Policy: Transmission channels

2 Main Transmission channels of Monetary Policy



(*): for sake of simplicity i is the interest rate determined by the Central Bank and it is used for discounting and borrowing in financial markets

Balance Sheet: Federal Reserve

2020, August (USD billion)

Assets

Liabilities and Net Worth

<u>Securities</u>	6,256*	Bank <u>reserves</u>	2,828
U.S. Treasury	4,320		
Agency MBS	1,934	<u>Currency</u> in circulation	1,957
Federal Agency Debt	2		
Repurchase agreements	0	U.S. Treasury Balance	1,635
		Reverse Repos	218
<u>Loans to FI and</u>		Other Liabilities	279
other liquidity facilities	83	Total Liabilities	6,918
Other assets	618		
Total Assets	6,957	Equity	39

* +2,359 since March, 2020

Federal Reserve Assets

Two main **asset** classes:

1. Securities

- before 2008 mostly *short*-term T-bills
- nowadays *long*-term Treasuries and Mortgage-Backed Securities (MBS) with QE (next slide)

2. Loans to Financial Institutions (discount window) or other liquidity facilities

- provisions of credit to help Financial Institutions and calm markets in crisis times
 - in normal times (and now) these are not used
 - but they were used a lot in crisis times (2008-2010)

Quantitative Easing

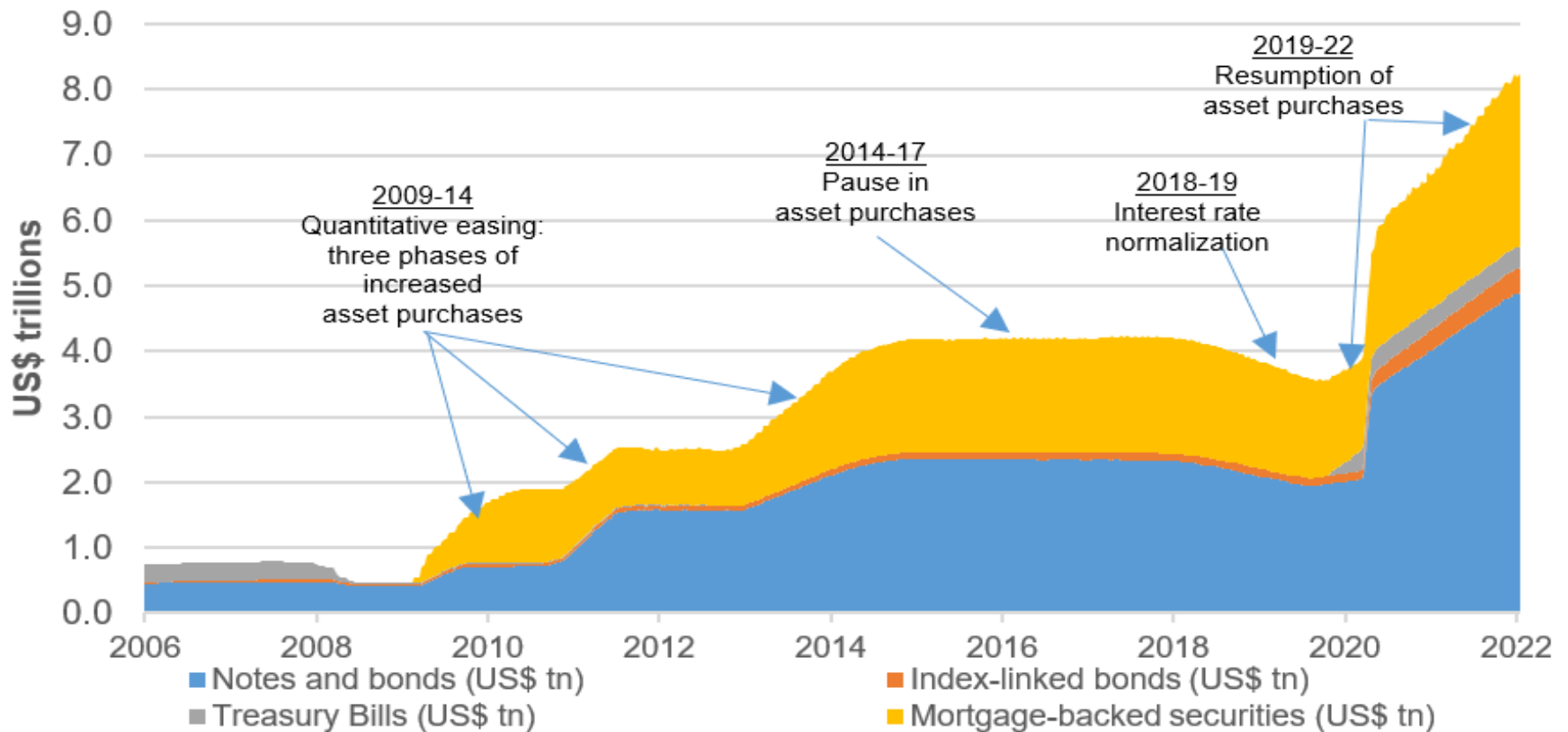
The Fed increased the amount of reserves to purchase securities and sustain the economy:

Quantitative Easing (QE)

- Nov 2008 QE1: **\$1.25** trillion in MBSs
- Nov 2010 QE2: **\$600** billion in Treasuries
- Sept 2012 QE3: **\$40** billion in MBSs each month until outlook improves
- Jan 2013 expanded Q3: additional **\$45** billion Treasuries
- March 2020: Covid

Quantitative Easing

Fed holdings of Treasury bonds and mortgage-backed securities during the quantitative easing era



Source: St. Louis Fed; RSM US

[More recent trends](#)

Federal Reserve Liabilities

Two main *liabilities*

1. Currency in circulation:

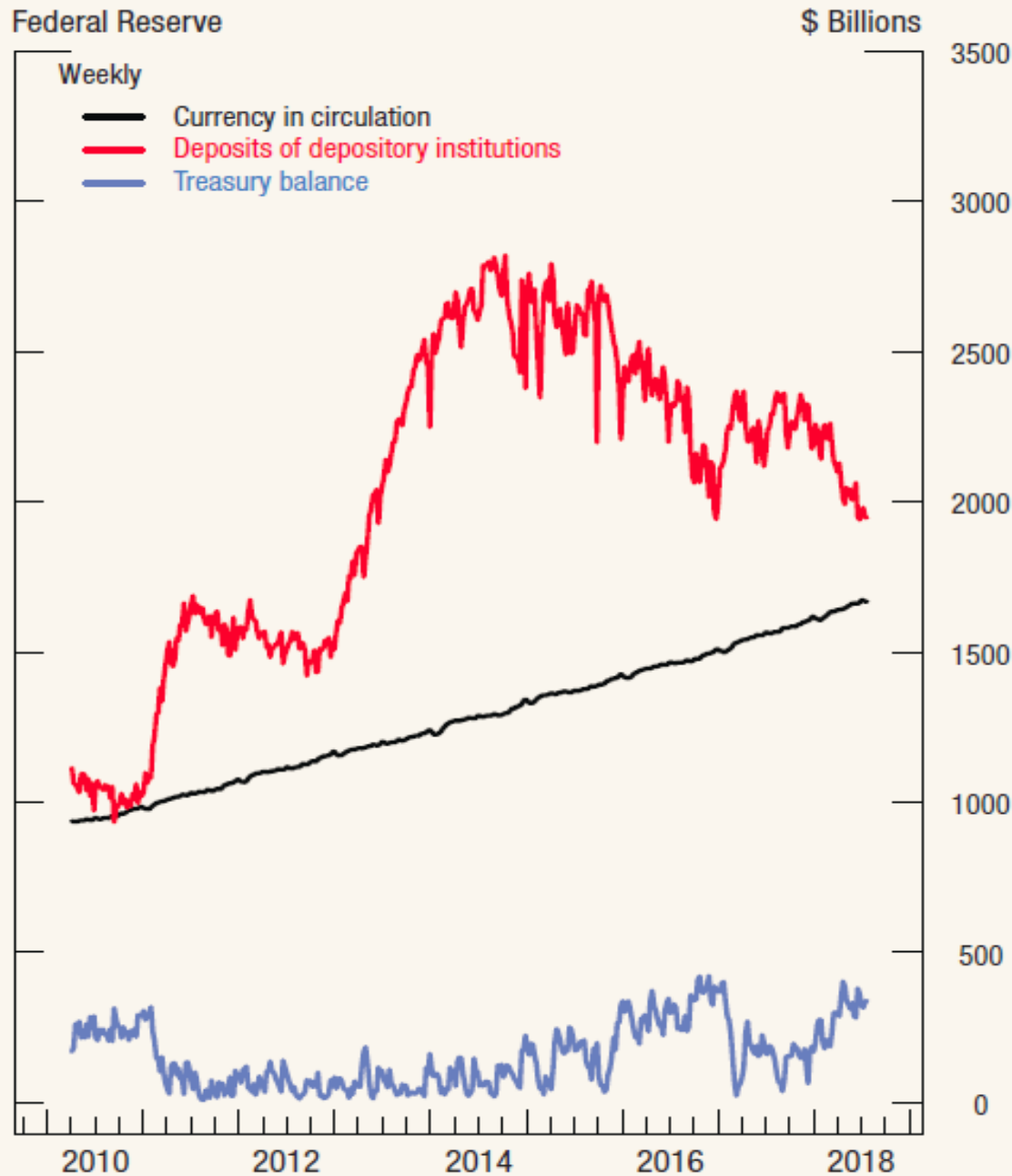
- banknotes and coins
- they grow at constant rate with economy

2. Reserves:

- funds banks hold in a *deposit account* at the Fed (Fed is like a bank for banks) + cash in bank vaults
- Fed *guarantees* that these deposits *can be redeemed* for physical cash *at any point in time*
- Two types:
 - **Required** reserves (RR): Fed sets the *required reserve ratio* ($rrr = \% \text{ checkable deposits}$) banks must hold in cash at Fed: minimum 3-10%
 - **Excess** reserves (ER): any reserves beyond RRs

Total Reserves = RR + ER

Selected Liabilities of the Federal Reserve



- *Currency* increases with economic growth
- Reserves increased much more after Financial Crisis of 2008

Source: Quarterly Report Fed Balance Sheet

Fed balance sheet over time

- ✓ The Fed balance sheet has expanded a lot in the last 10 years, after each crisis (2008 and 2020)
- ✓ But the basic structure of
 - assets (securities + loans to banks)
and
 - liabilities (currency in circulation + reserves)
... has not changed
- ✓ ... just the scale (and type of securities held) is different
- ✓ It is useful to start from what the Fed *used* to do in normal times to conduct monetary policy:
open market operations (conventional MP instruments)

Normal times: open market operations

- ✓ Open market operations refer to either:
 1. *purchase/sale* of securities (T-bills) by the Fed from/to a set of banks (with **primary market dealers**)
 2. *discount window lending* to financial institutions
- ✓ The objective is to influence the amount of **money supply** (i.e. of **bank reserves**)
 - *purchase/sale* of securities: reserves ↑/↓
 - *borrowing/repaying from/to Fed*: reserves ↑/↓
 - so (mostly by trading securities), Fed changes amount of reserves in the banking system
- ✓ Bank reserves can be exchanged by banks in the **Federal Funds market** (a *secondary money market*)
 - which means *safe*, short-term (overnight) instrument

The Federal Reserve Balance Sheet

1. *Purchase of securities* (“*Non-Borrowed Reserves: NBR*”)

- ✓ Suppose the Fed wants to purchase \$100m securities

The Fed	
Assets	Liabilities
Securities +\$100 m	

- ✓ How does it pay for it?

The Federal Reserve Balance Sheet

1. *Purchase of securities* (“*Non-Borrowed Reserves: NBR*”)

- ✓ Suppose the Fed wants to purchase \$100m securities

The Fed	
Assets	Liabilities
Securities	Reserves
+\$100 m	+\$100 m

- ✓ How does it pay for it? With **Reserves**
 - Fed can just issue reserves: newly created reserves for free (recall monetary expansion)
 - the Central Bank (Fed here) is the only bank that can do it!
 - historically, purchased securities were almost exclusively T-Bills

The Federal Reserve Balance Sheet

✓ If Fed buys securities from a bank, then:

The Fed	
Assets	Liabilities
Securities +\$100 m	Reserves +\$100 m

if balance sheet
of the FED
expands ...

Banking System	
Assets	Liabilities

The Federal Reserve Balance Sheet

✓ If Fed buys securities from a bank, then:

The Fed	
Assets	Liabilities
Securities +\$100 m	Reserves +\$100 m

balance sheet
of the FED
expands

Banking System	
Assets	Liabilities
Securities -\$100 m	
Reserves +\$100 m	

... bank balance
sheet size is
constant,
but there are
more reserves

The Federal Reserve Balance Sheet

✓ If Fed buys securities from the public (i.e. us)?

The Fed

Assets	Liabilities
Securities +\$100 m	Reserves +\$100 m

Banking System

Assets	Liabilities

The Public

Assets	Liabilities

The Federal Reserve Balance Sheet

- ✓ If Fed buys securities from the public (i.e. us)?

The Fed

Assets	Liabilities
Securities +\$100 m	Reserves +\$100 m

Banking System

Assets	Liabilities
Reserves +\$100m	Deposits +\$100m

The Public

Assets	Liabilities
Securities -\$100 m	
Deposits +\$100m	

- the public sells securities to Fed, and receives a credit on her checking account (Deposits at bank ↑)
- the bank increases Deposits, and receives Reserves from Fed
- note: *bank increases reserves and balance sheet*

The Federal Reserve Balance Sheet

2. *Discount Lending (“Borrowed Reserves: BR”)*

- ✓ Suppose the Fed lends \$100M to a bank

The Fed		Banking System	
Assets	Liabilities	Assets	Liabilities
Discount Loans	Reserves	Reserves	Borrowed Reserves
+\$100 M	+\$100 M	+\$100 M	+\$100 M

- ✓ Discount loans **expand** liquidity in the banking system by increasing reserves that can be lent out to the economy:
 - assets: reserves go **up** by \$100 M
 - liabilities: borrowing from FED go **up** by \$100 M
- ✓ Fed charges banks high interest:
 - the “**discount rate**” is usually around **100 bps** (=1 percentage point) **above** the interbank rate (=Fed funds rate)
 - discount loans are not much in demand in normal times, but important in *financial panics*: Fed as ***lender of last resort***

Supply and Demand in the Market for Reserves



Now examine how change in reserves affects the **Federal funds rate**

- the rate banks charge each other for *overnight* loans



Further, we examine other tools available to the Fed

-the ability to set the ***required reserve ratio (rrr)*** for deposits held by banks



We will also see how this market has changed after the 2008 crisis

Supply and Demand in the Market for Reserves

There are 3 rates to keep in mind:

1. the **discount window rate** (i_d)
 - this is the rate at which banks can borrow directly from the Fed
unlimited quantities of reserves
2. the **Fed funds target rate** (i_{ff})
 - the *effective* rate at which banks lend and borrow *from each other* (equilibrium) may be different from *target* from time to time, but very close
 - this is the rate the Fed “*wants*” to see for **interbank** borrowing
3. the **interest rate on Excess Reserves** (i_{OR})
 - this is the rate the Fed pays on excess reserves *deposited* at Fed
(*Fed is borrower*)

Supply and Demand in the Market for Reserves

i (Federal Funds Rate)

$i > i_d$: No!

- banks would borrow from Fed and lend to other banks at higher rate
- no bank would borrow from other banks, just from Fed

Discount rate sets a **ceiling**

→ i_d

Fed Funds target rate

→ i_{ff}

Interest on **excess reserves** sets a **floor**

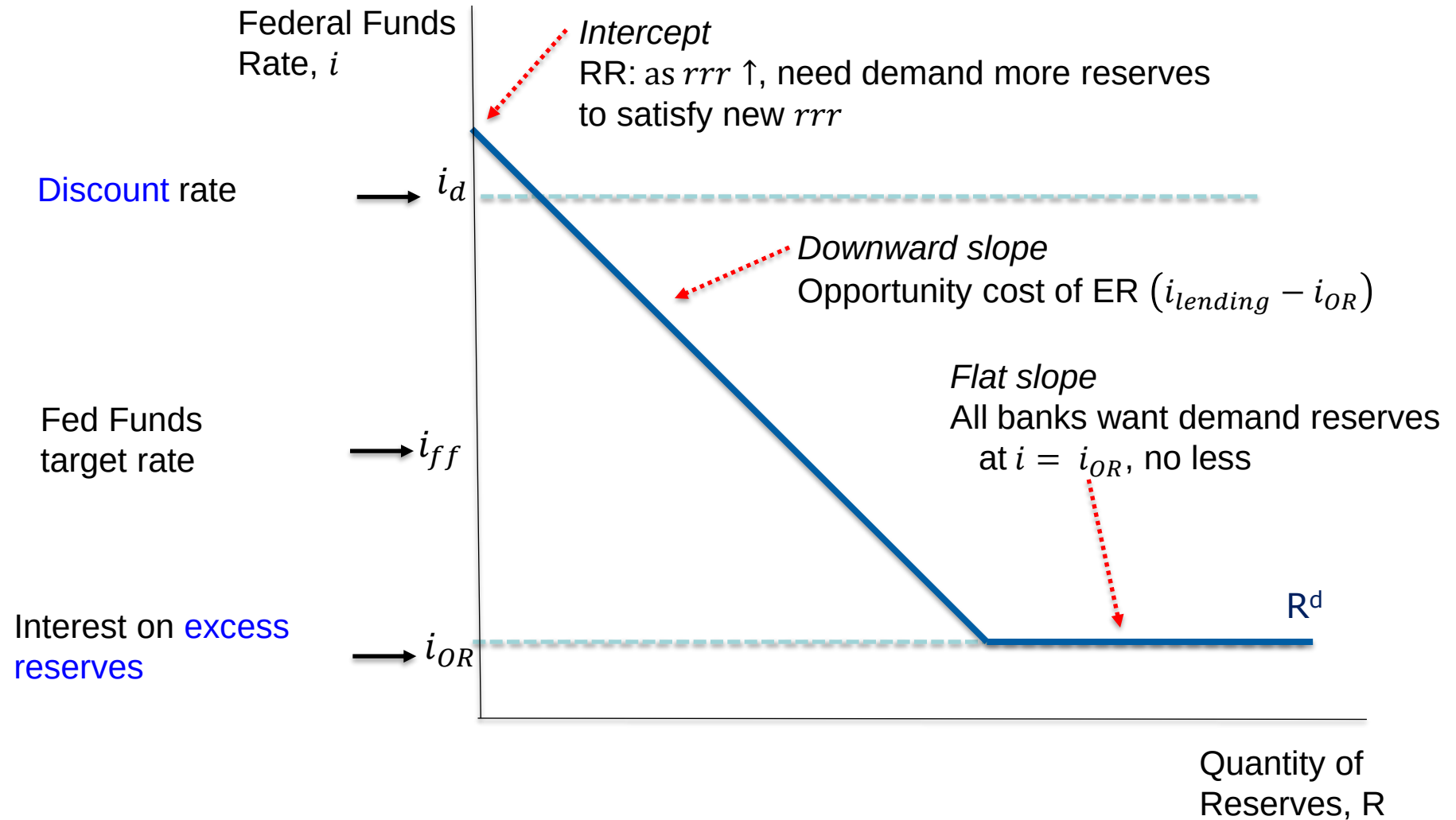
→ i_{OR}

$i < i_{OR}$: No!

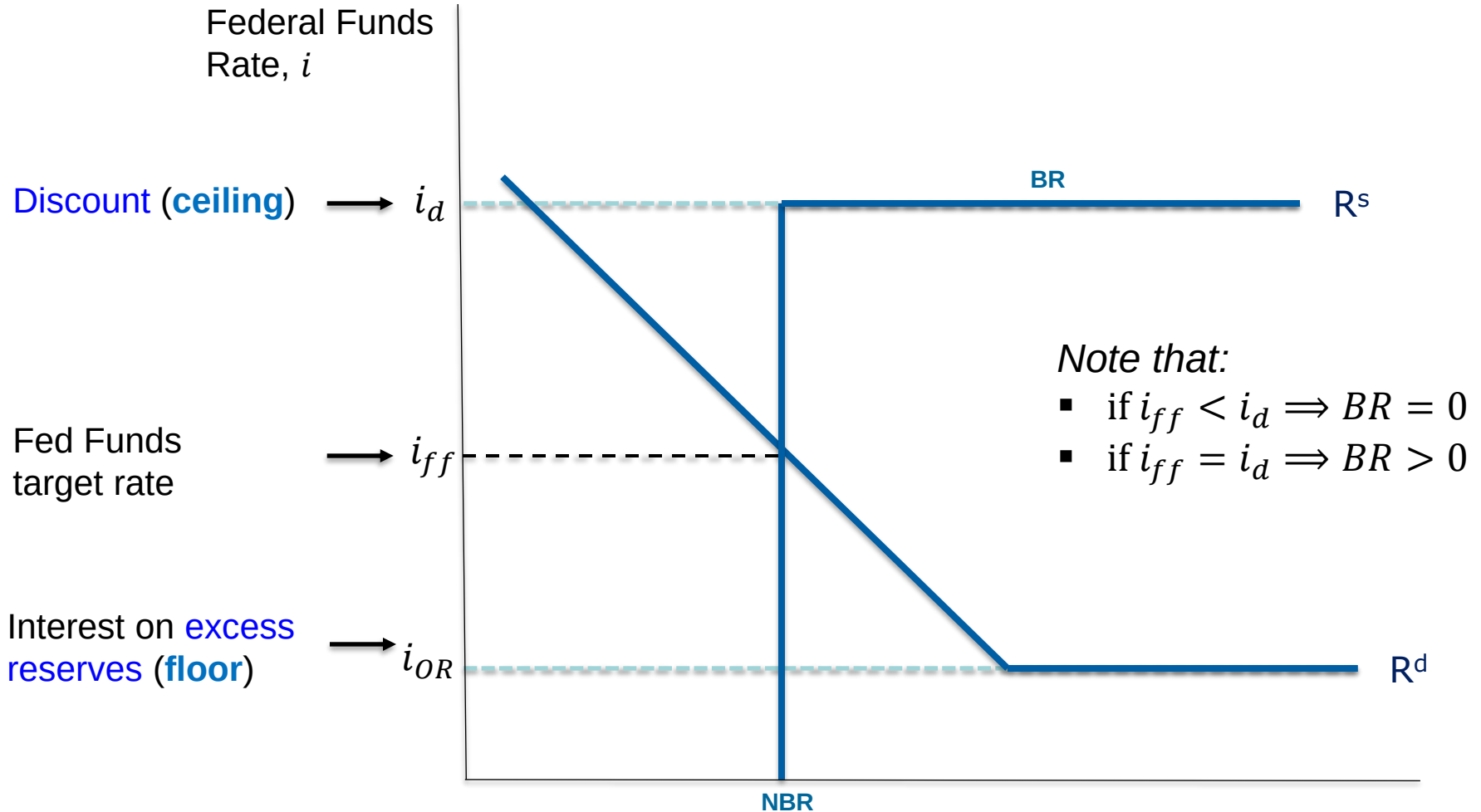
- banks would borrow from other banks and then deposit reserves at Fed
- no bank would lend to other banks (**worse borrower** than Fed) at a lower rate, just use deposit facility

Quantity of Reserves, R

Demand of Reserves

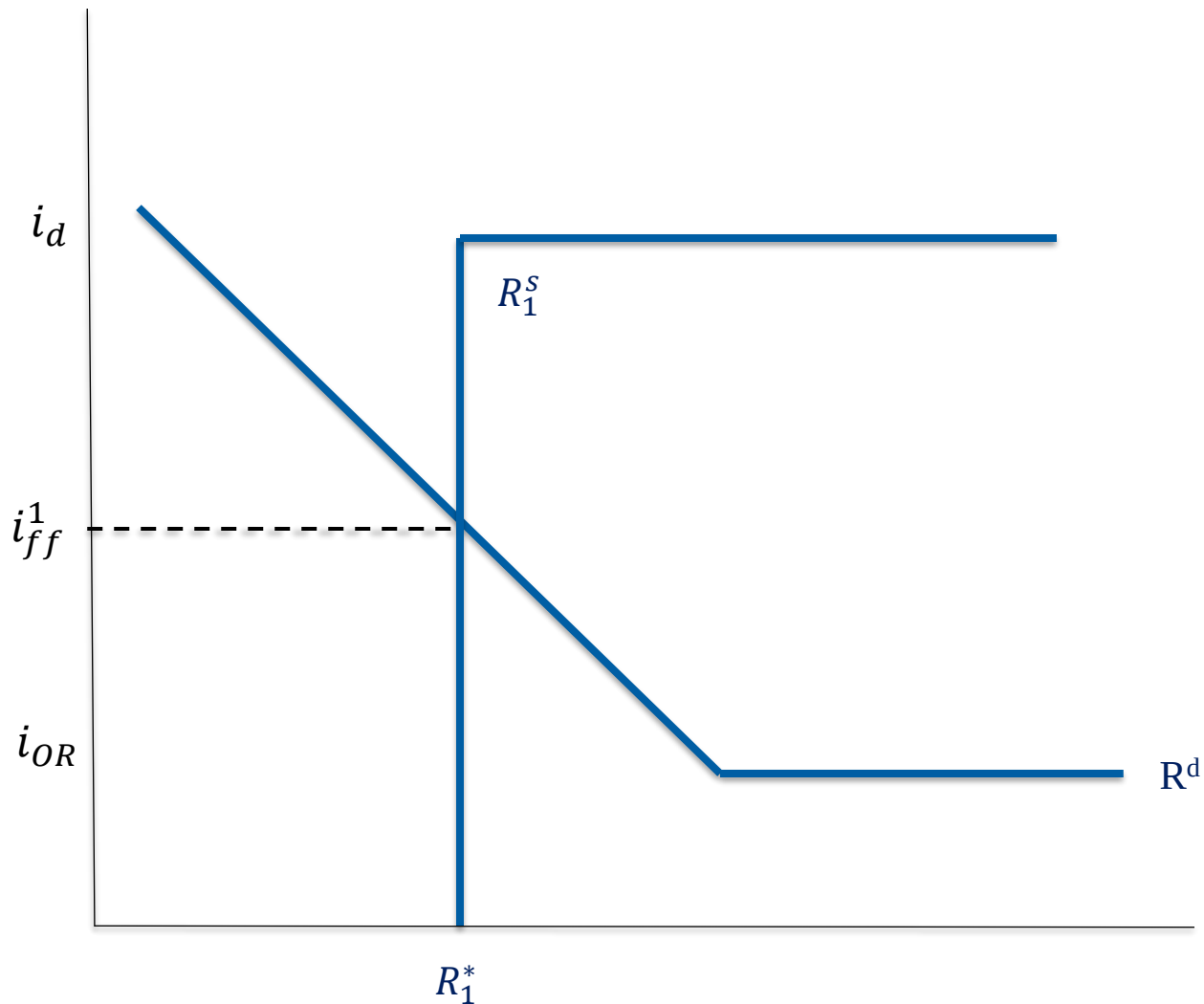


Supply and Demand in the Market for Reserves

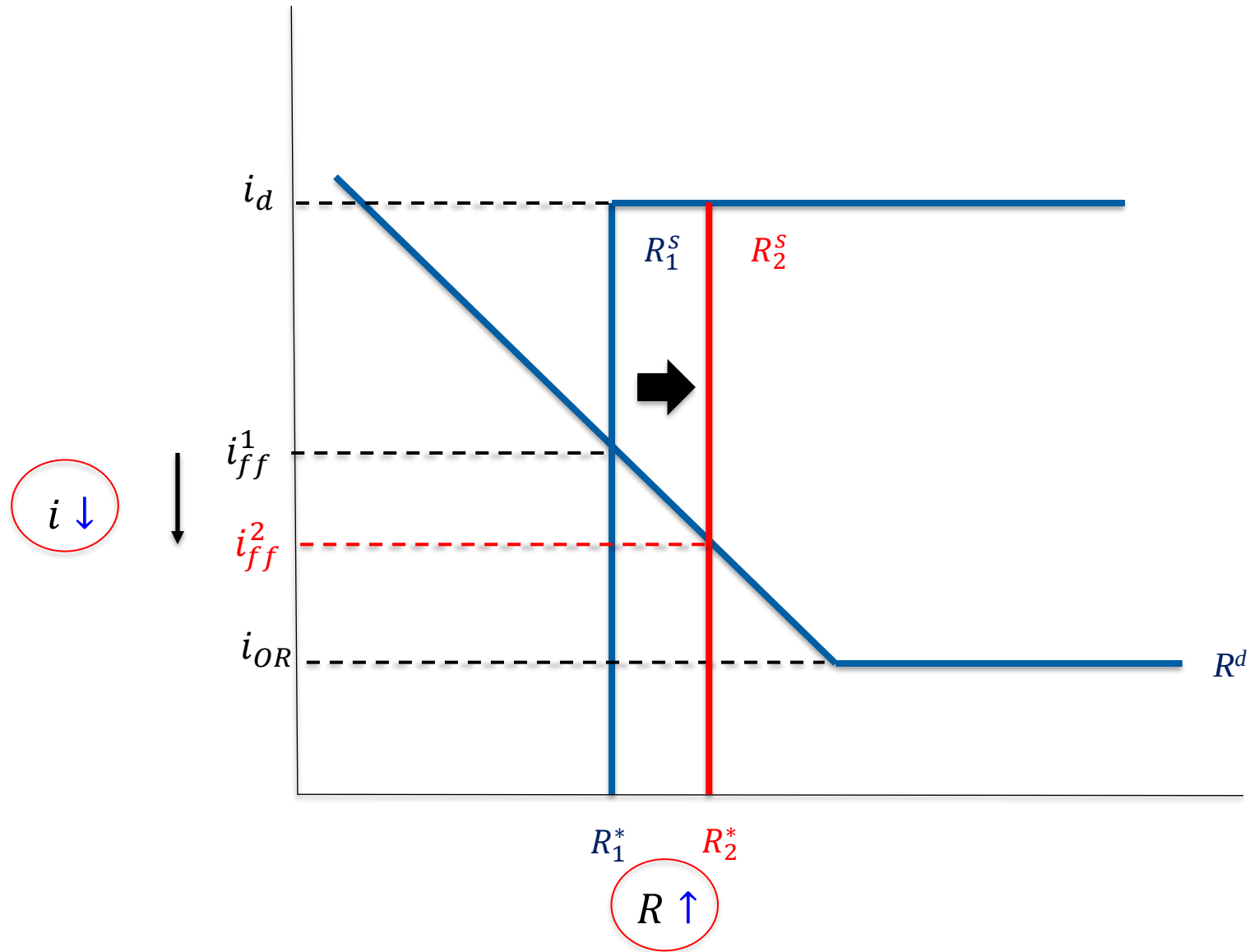


Supply curve is vertical because Fed is monopolist in market for Reserves, but at $i = i_d$, it supplies unlimited reserves (discount window borrowing)

Response to Open Market Purchase



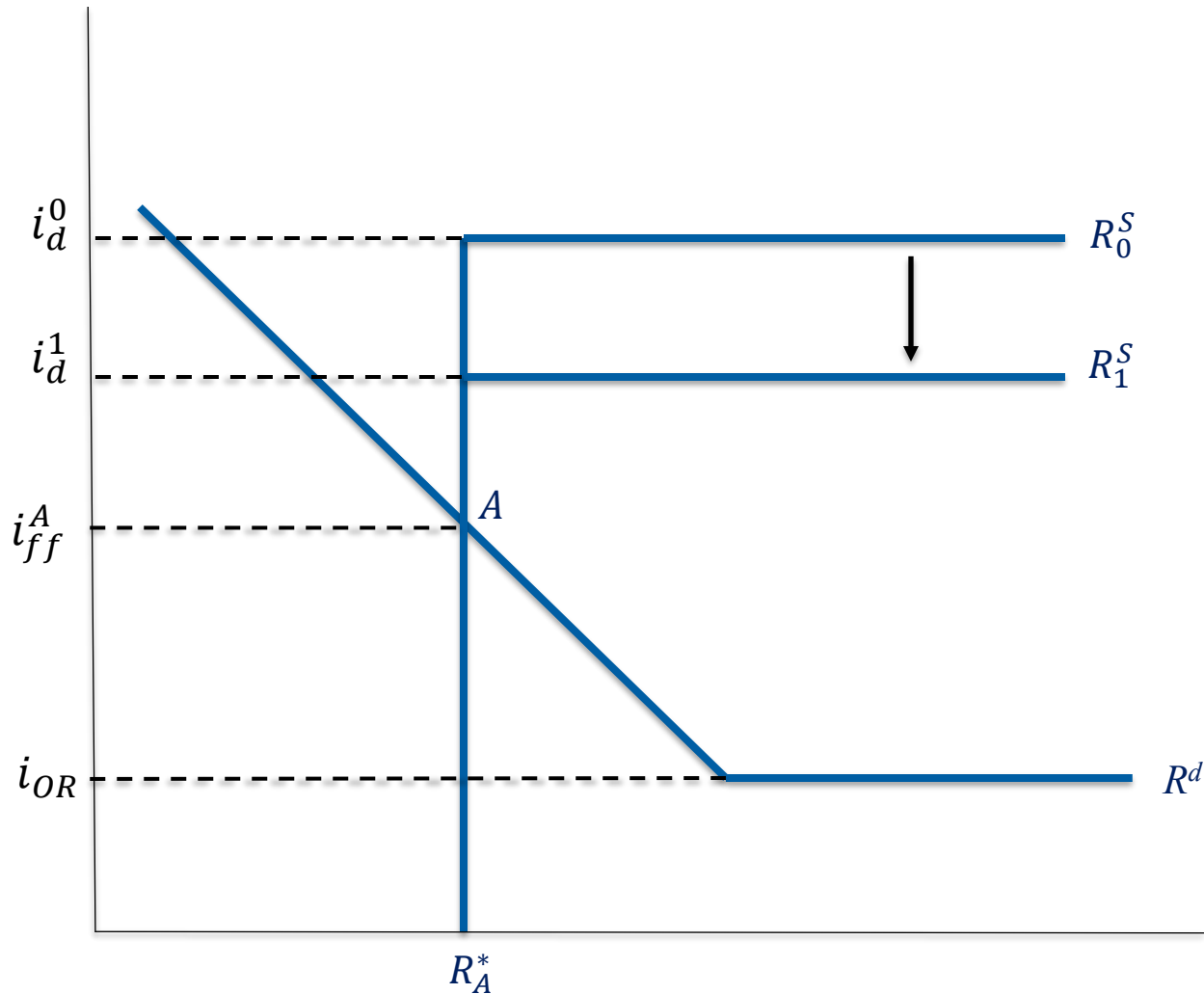
Response to Open Market Purchase



Response to Open Market Purchase

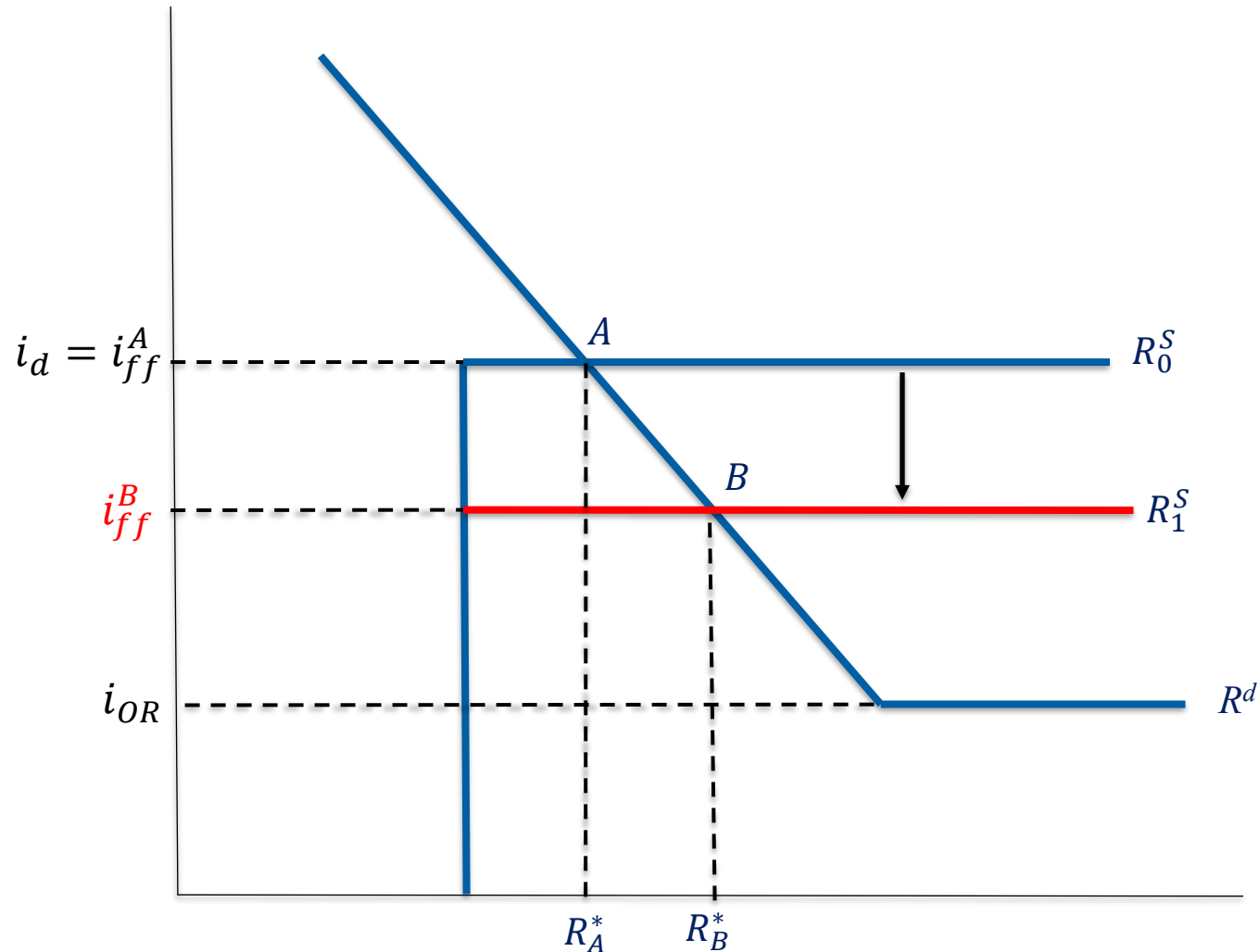
- ✓ With open market purchases / sales
 - Reserves ↑ / ↓, Fed Fund target rate ↓ / ↑
- ✓ For this to work, the supply curve needs to intersect the demand curve on its downward sloping part
- ✓ This is what happens in **normal times** when reserves are **scarce**

Lowering Discount rate (typical case)



Typically Fed wants $i_d > i_{ff}$ (supply intersects demand in downward sloping part):
changing i_d **does not** affect i_{ff}

Lowering Discount rate (particular case)

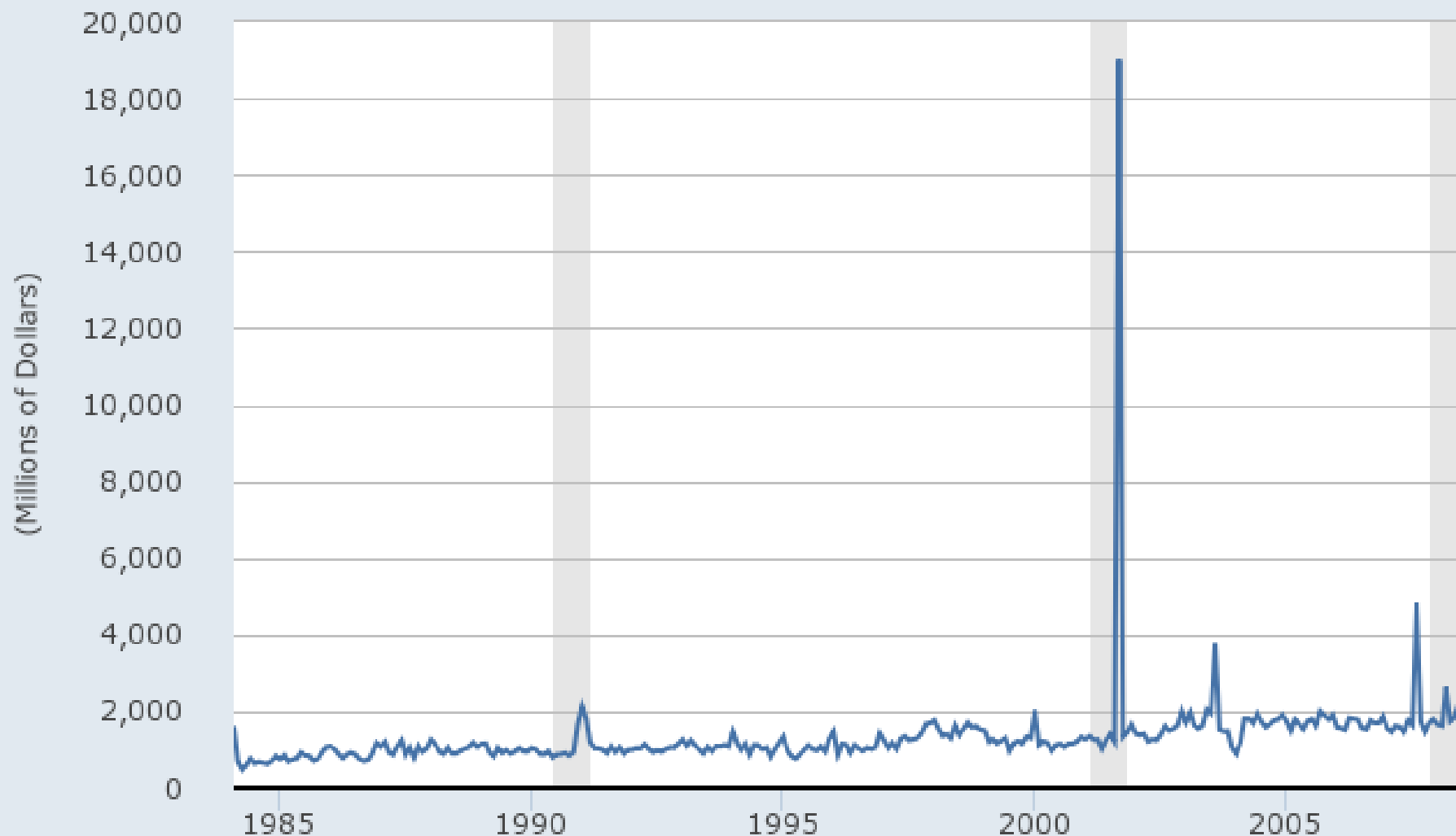


Particular case: $i_d = i_{ff}$ (with supply intersecting demand in flat sloping part):
changing i_d **affects** i_{ff}

Excess Reserves, 1985-2008



— Excess Reserves of Depository Institutions



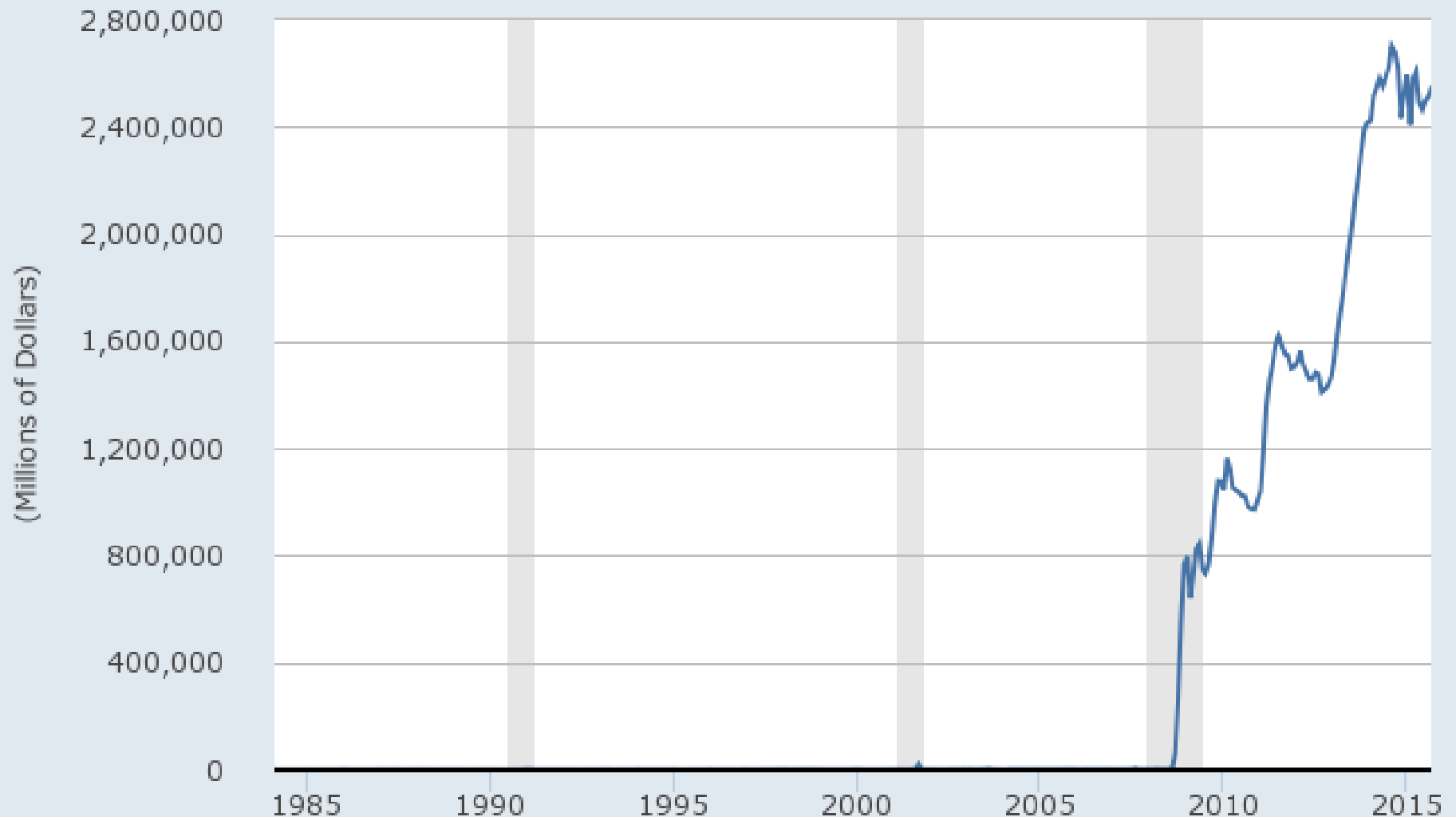
Source: Federal Reserve Bank of St. Louis

Shaded areas indicate US recessions - 2015 research.stlouisfed.org

Excess Reserves, 1985-2015



— Excess Reserves of Depository Institutions



Source: Federal Reserve Bank of St. Louis

Shaded areas indicate US recessions - 2015 research.stlouisfed.org

Excess Reserves, 1985-2020



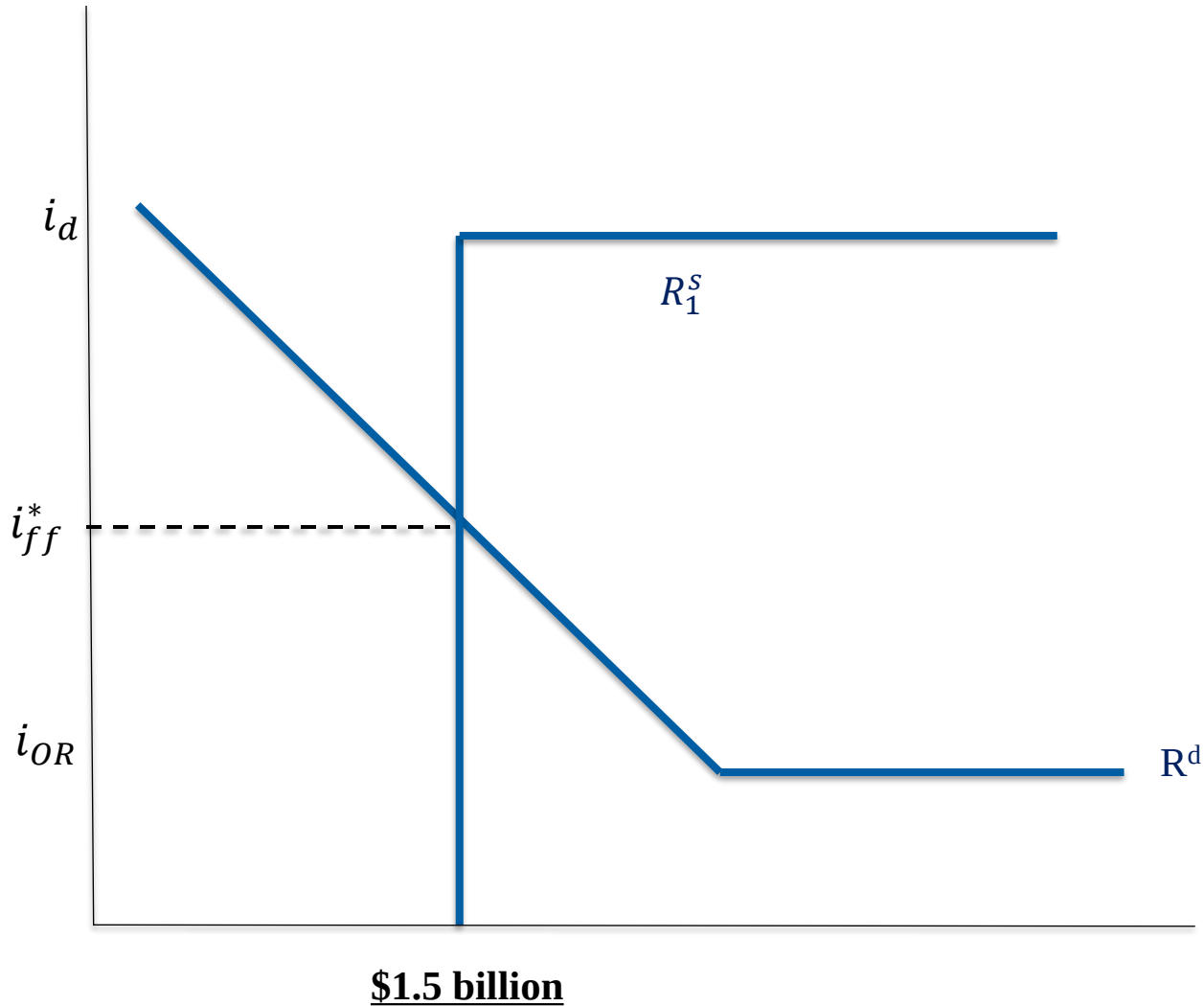
Shaded areas indicate U.S. recessions.

Source: Federal Reserve Bank of St. Louis

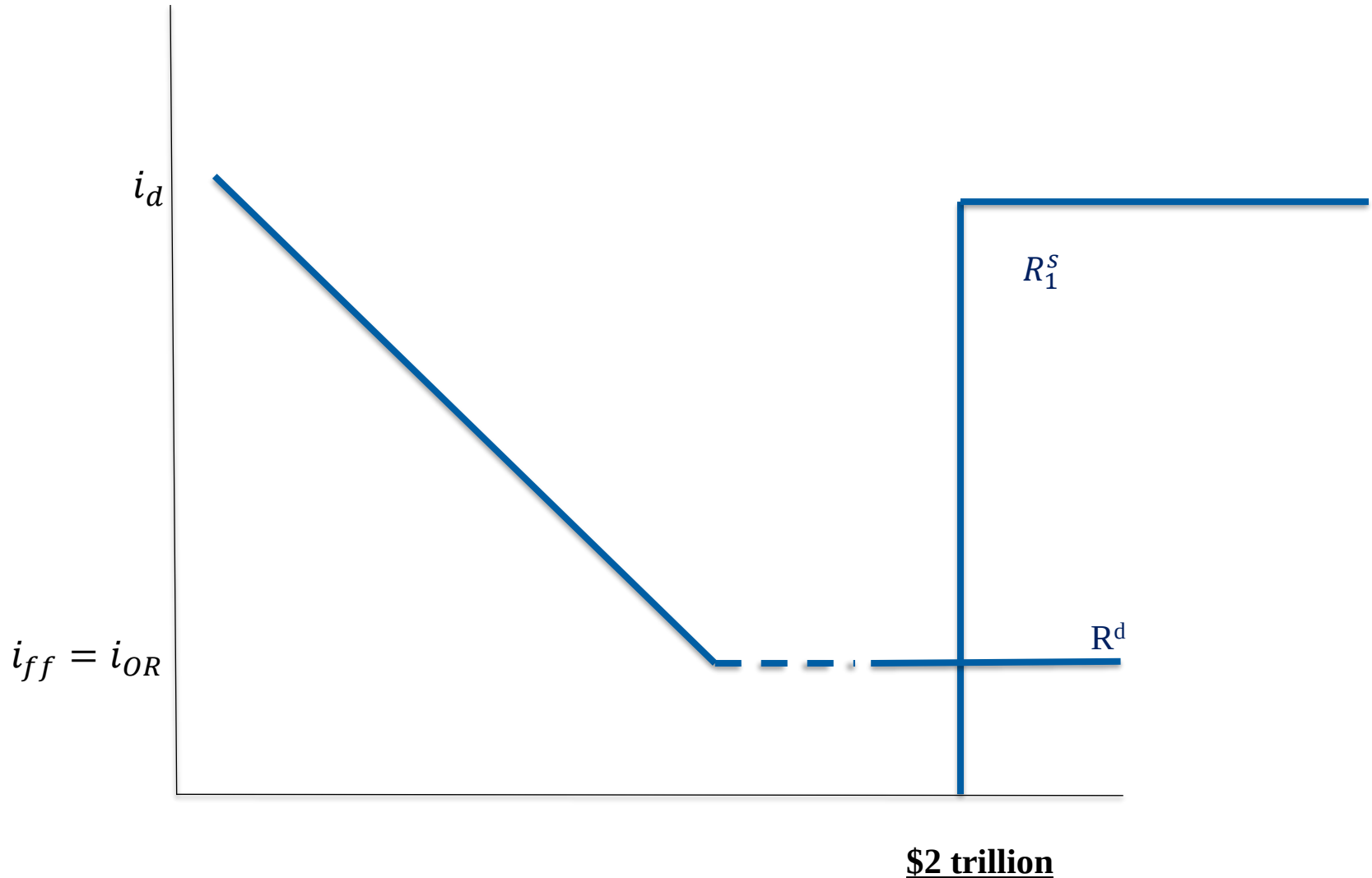
fred.stlouisfed.org



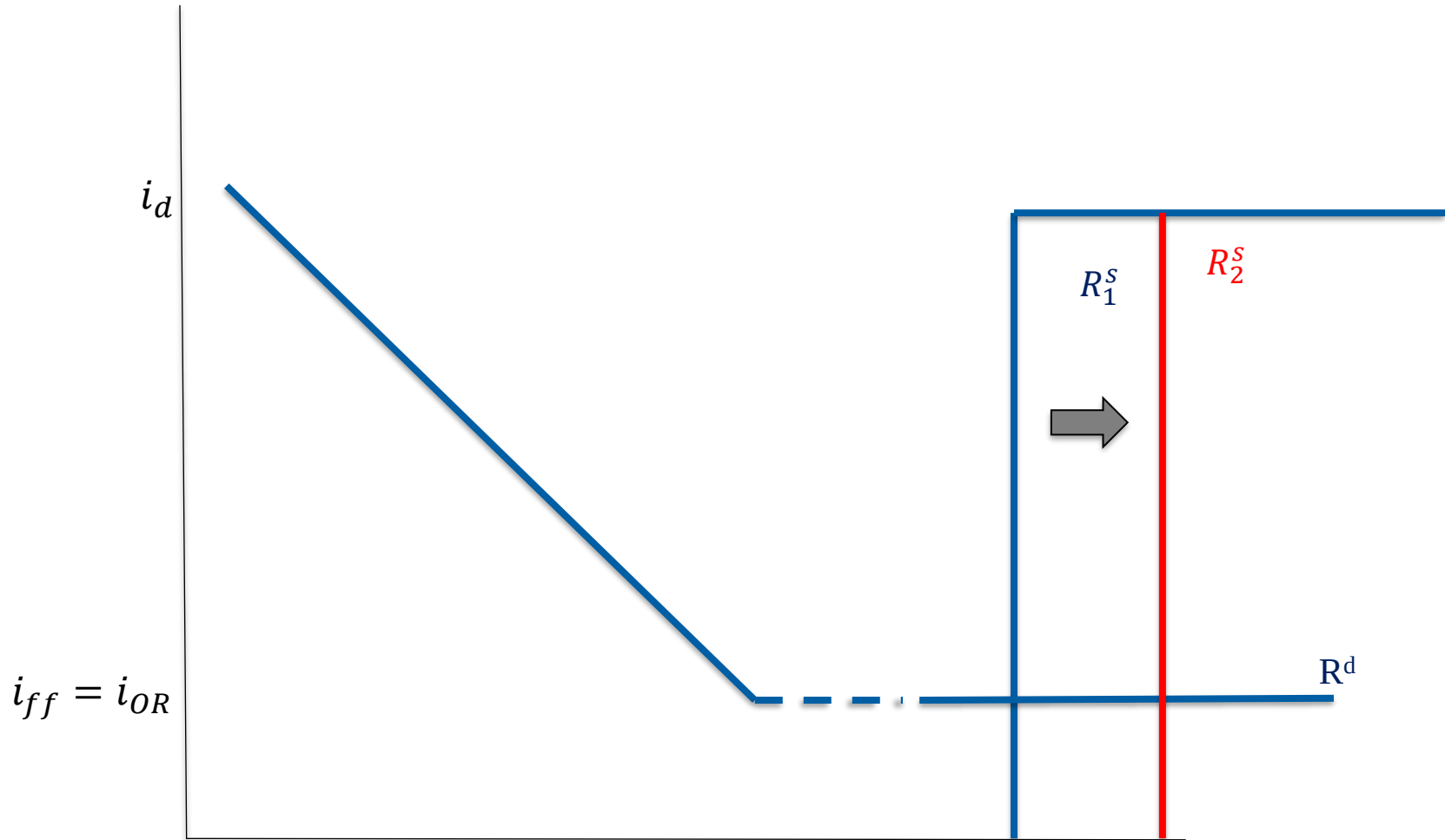
Reserves Scarcity (<2008)



Reserves Super-Abundant (“yesterday”)



Response to Open Market Purchase with Super-Abundant reserves (“yesterday”)



- Open market operations are ineffective on i \$2 trillion R_2^*
- Equilibrium Funds rate **remains** at i_{OR}

Monetary Policy nowadays



with abundance of reserves

open market operations do **not** work

in fact the the Fed funds rate has been close to zero from 2008 to 2016

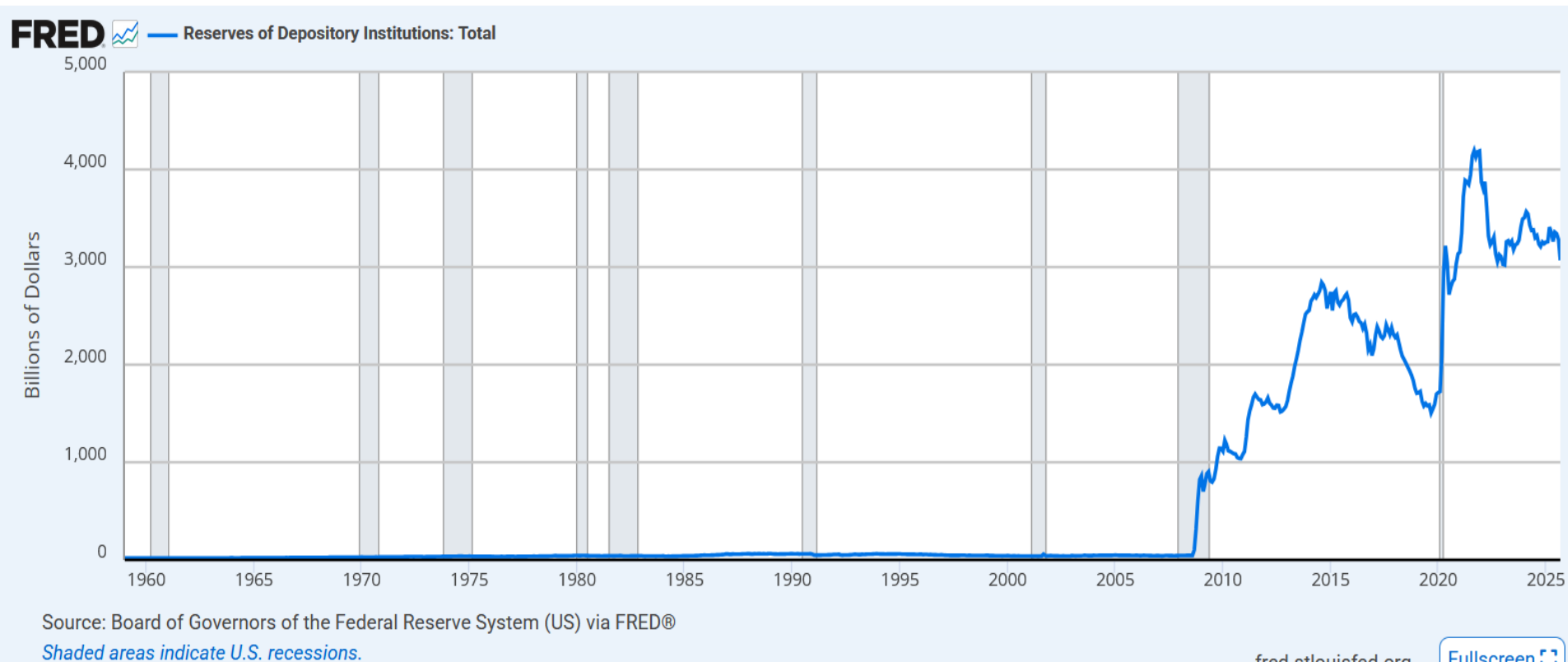


«Today» rates are high, and the Fed balance sheet is **shrinking** (*Quantitative Tightening, QT*)

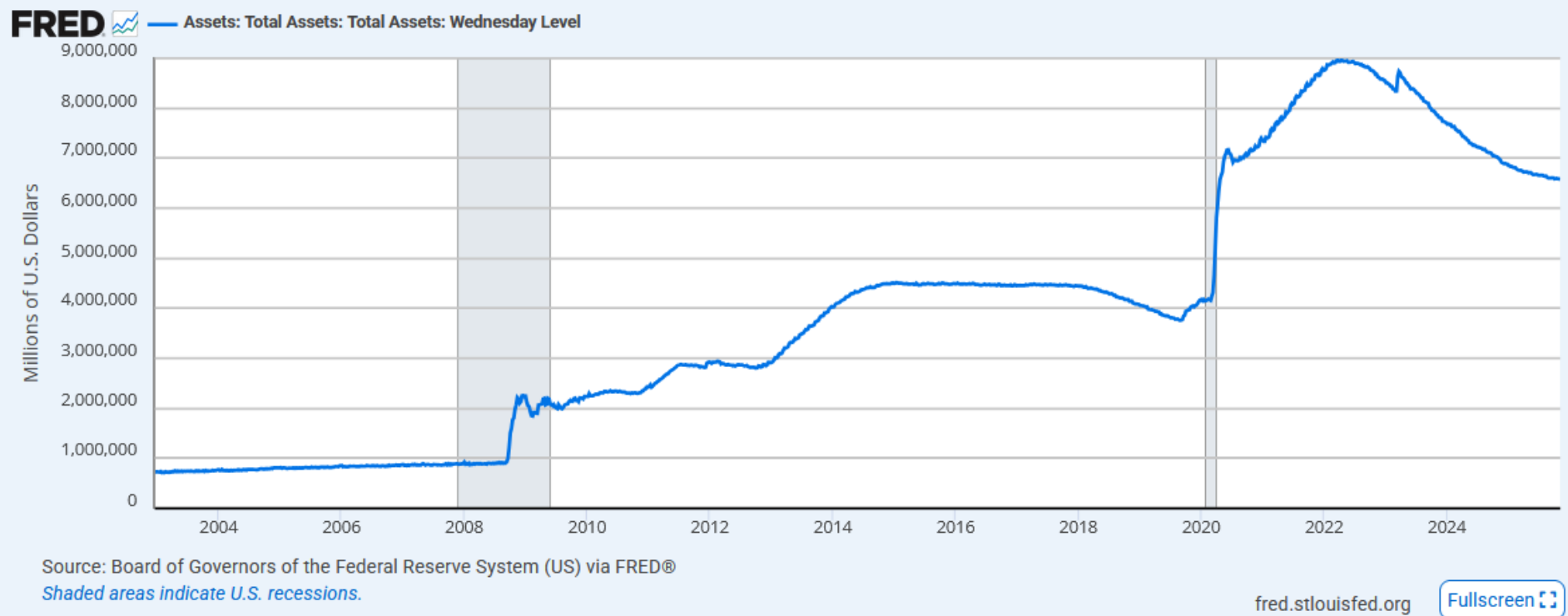
it will take a years before both reserves and securities are back to pre-2008 level

(perhaps never going back to that world)

Slow Quantitative Tightening (1/2)



Slow Quantitative Tightening (2/2)

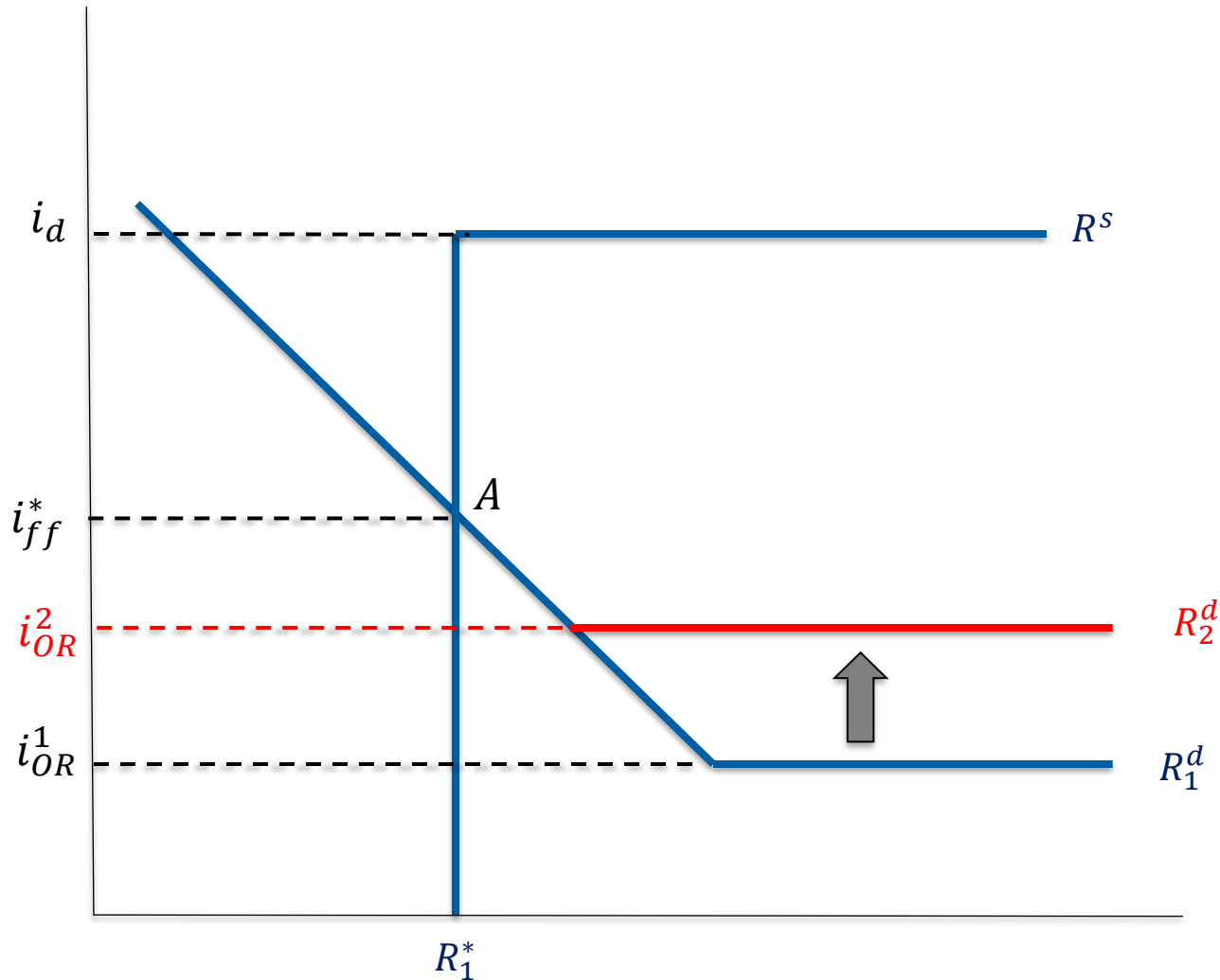


Monetary Policy: other tools

The Fed has several tools at its disposal to increase Fed funds rates even with large reserves:

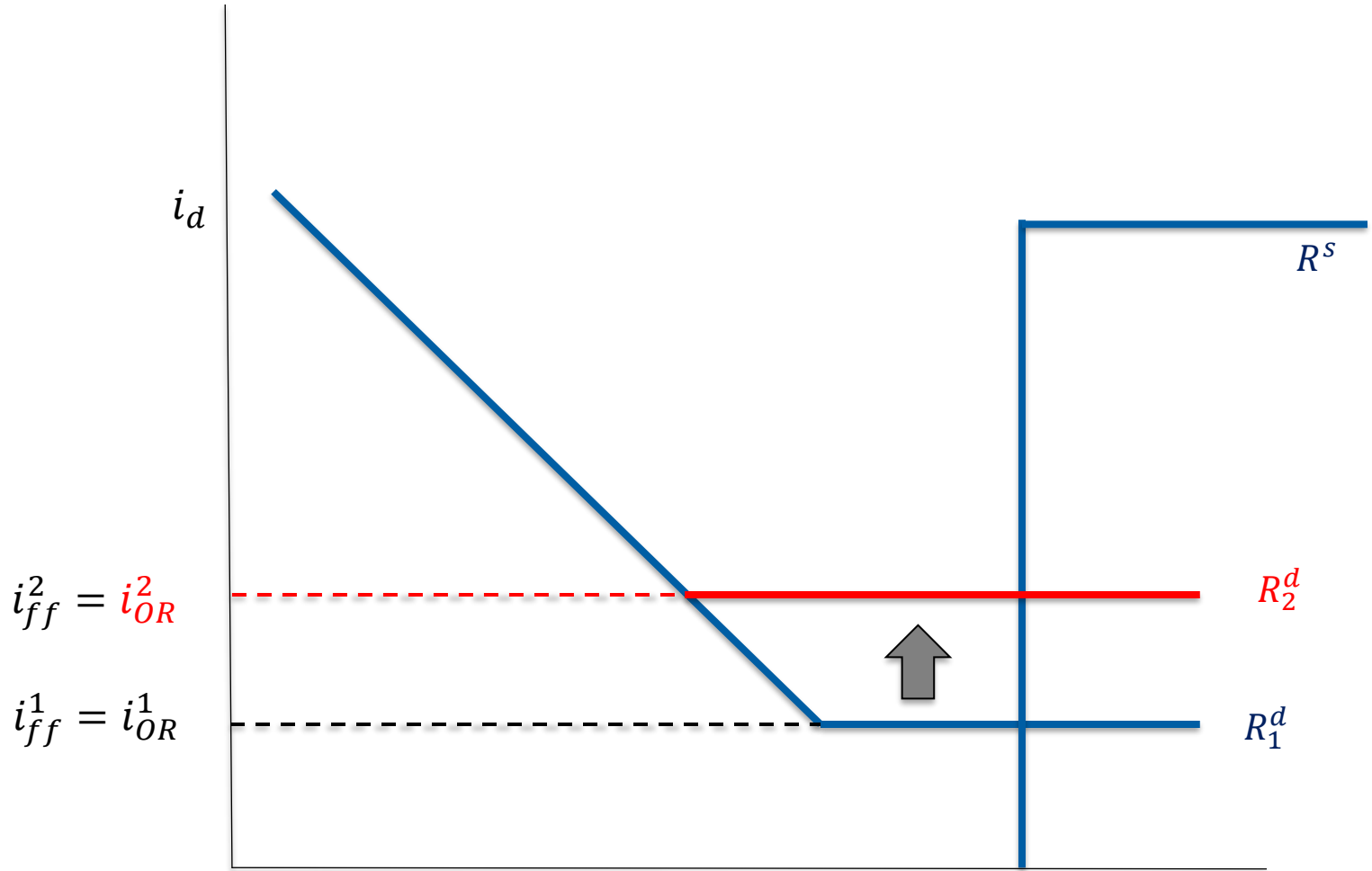
- 1. Increase interest on excess reserves (i_{OR})**
- 2. Increase reserve requirements ratio (rrr)**
- 3. Reverse Repos (more technical)**

Increase Interest on Excess Reserves



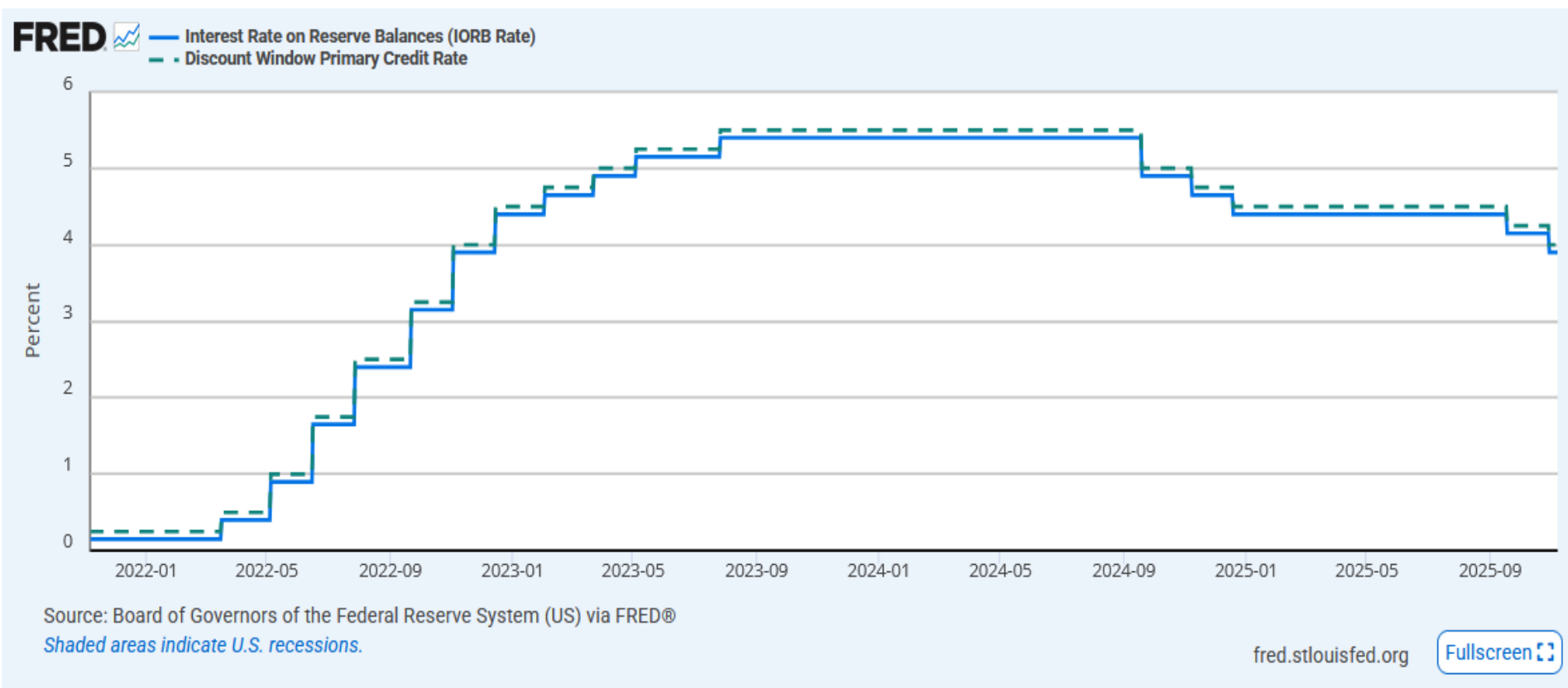
- In the **pre**-crisis world (with supply intersecting demand in downward sloping part): changing i_{OR} **unlikely to affect** i_{ff}

Increase Interest on Reserves

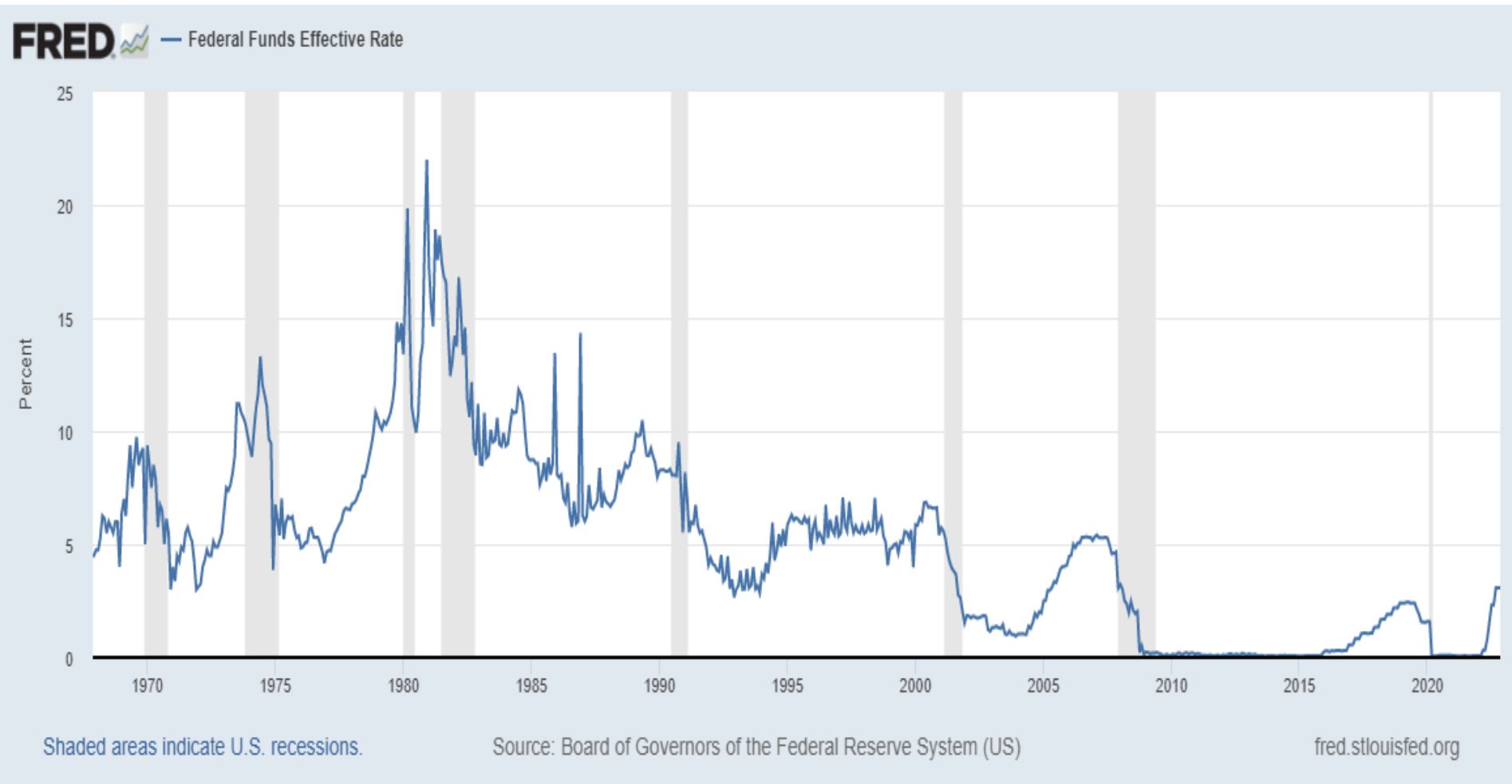


- But in **post** crisis world, since R^s intersects the demand curve far to the right: changing i_{OR} **does affect** Fed Funds rate

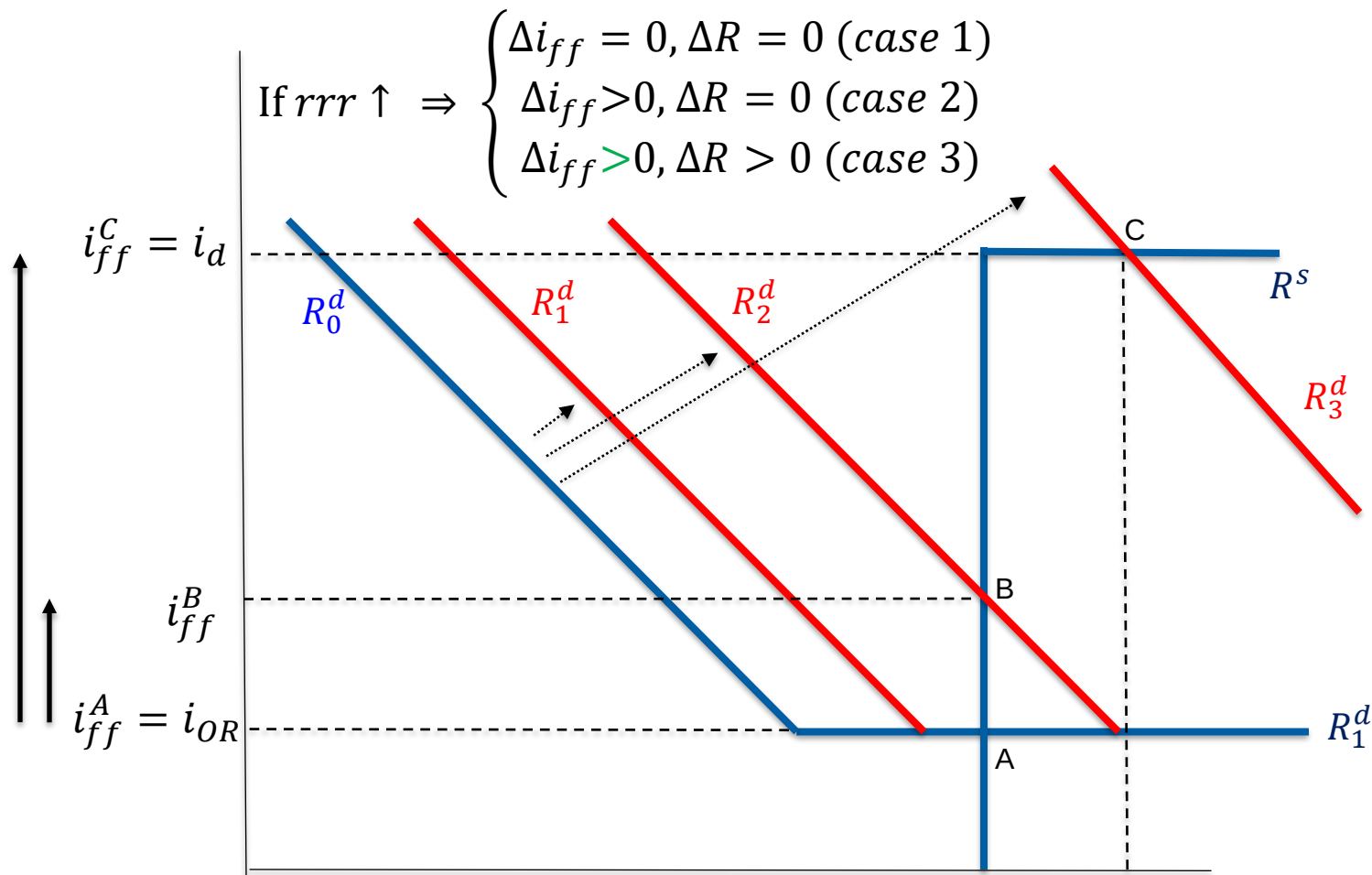
Fed Funds Rate: Discount and Excess Reserves



Historically high, not that much today



Increase in Required Reserves



- case 1: *particular* case (**abundant** reserves), when flat part of demand meets vertical part of supply*
- case 2: *normal* case (reserve **scarcity**), when downward part of demand meets vertical part of supply (as in figure)*
- case 3: *rarest* case (theoretical), when downward part of demand meets flat part of supply*

*: all specifications refer to final equilibrium.

Other Central Banks

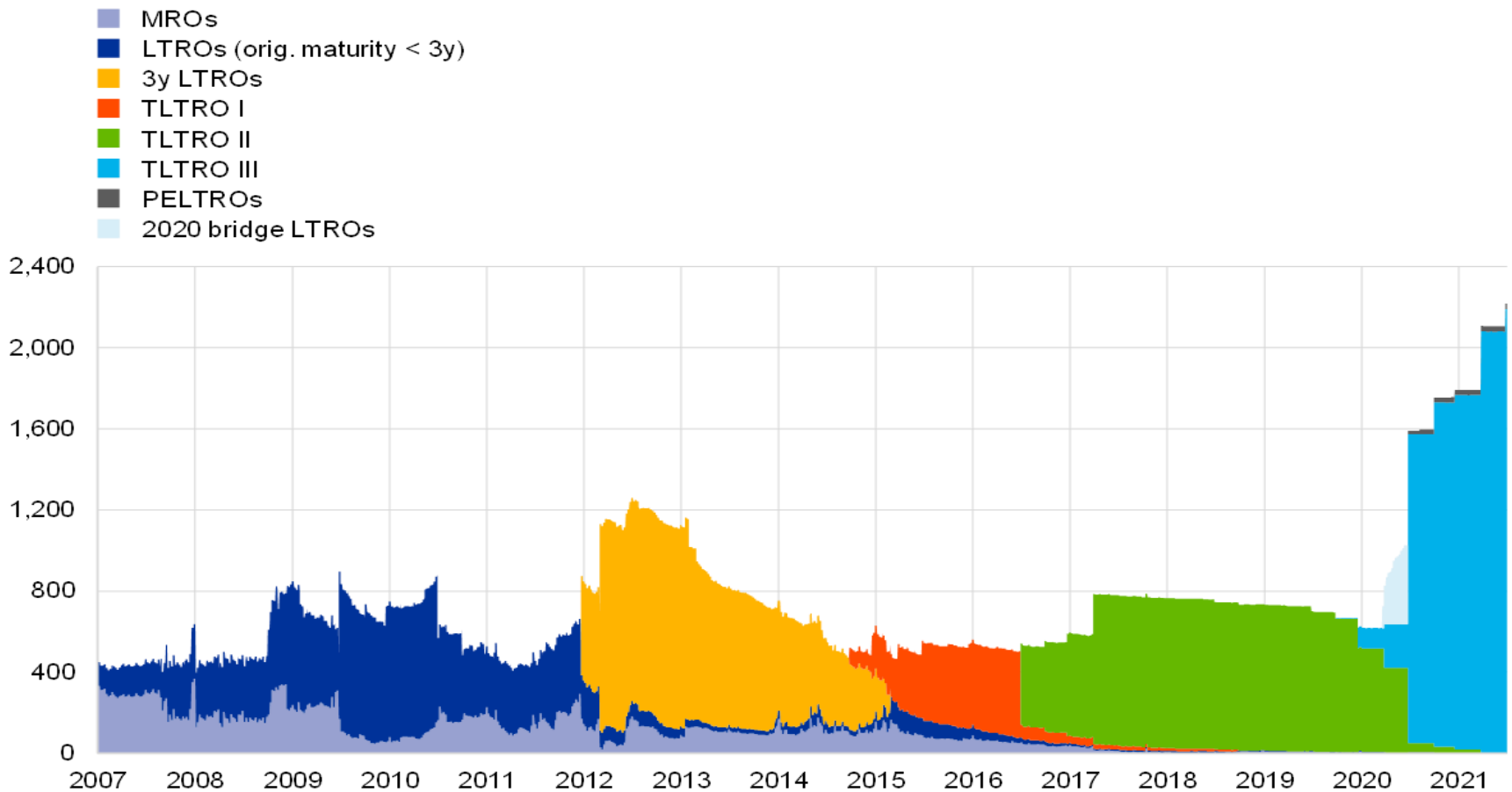
The EBC has similar tools to implement its policy:

- (1) *open market operations*: **main refinancing operations** (like repos) via a bid system from its credit institutions (competitive bids before 2008, uncompetitive after)
- (2) lending to banks: *marginal lending facility* (ceiling of o/n int. rate)
- (3) *deposit facility* (floor of o/n int. rate)
- (4) reserve requirements

ECB operations are decentralized

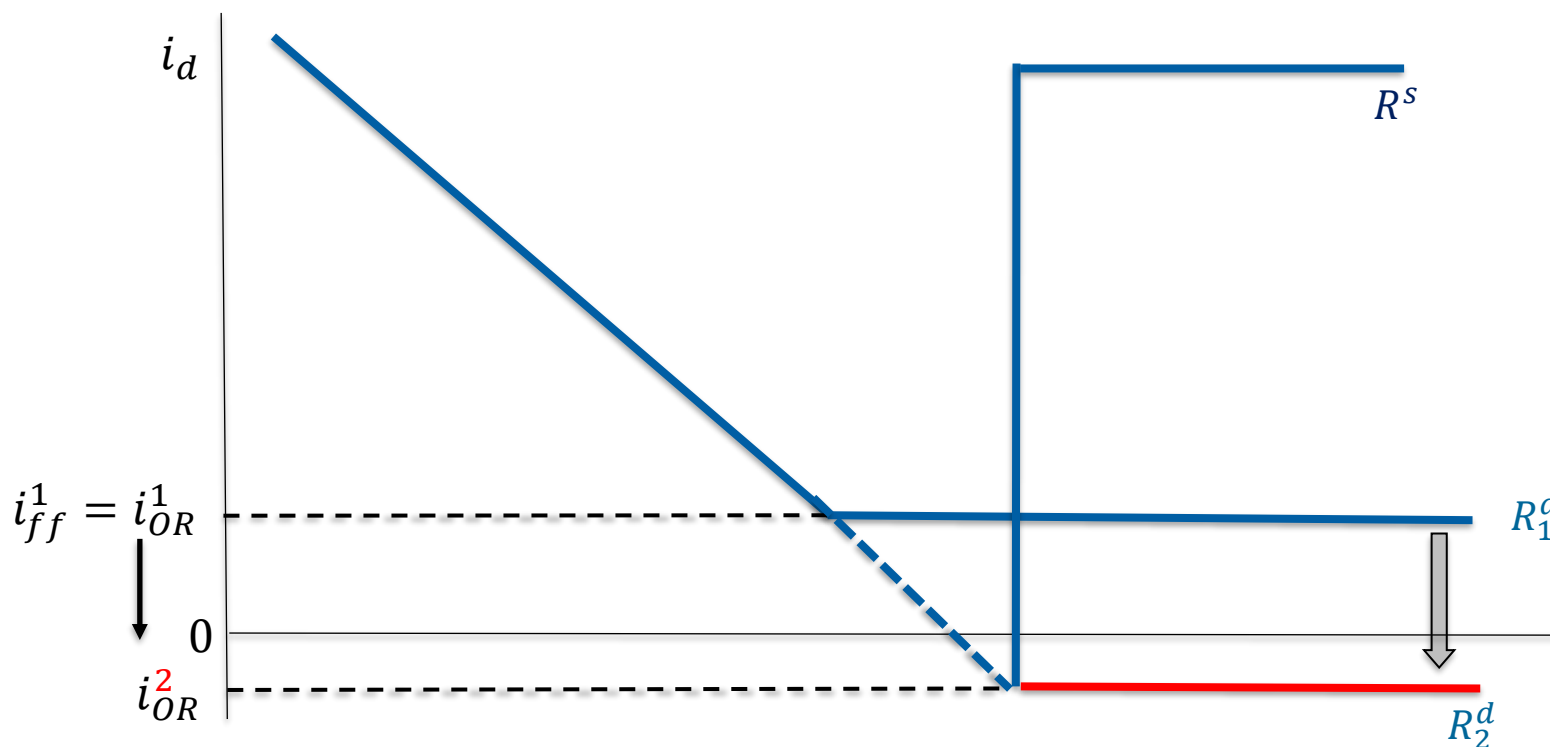
- carried out by each nation's central bank

- ✓ Until recently, all borrowing from the ECB was composed of funds borrowed at **negative rates** (deposit facility) for three years, with the agreement that funds *have to be used for new loans* (*Targeted Long Term Refinancing Operations*, TLTRO's)
- ✓ Now deposit facility rate is 2%

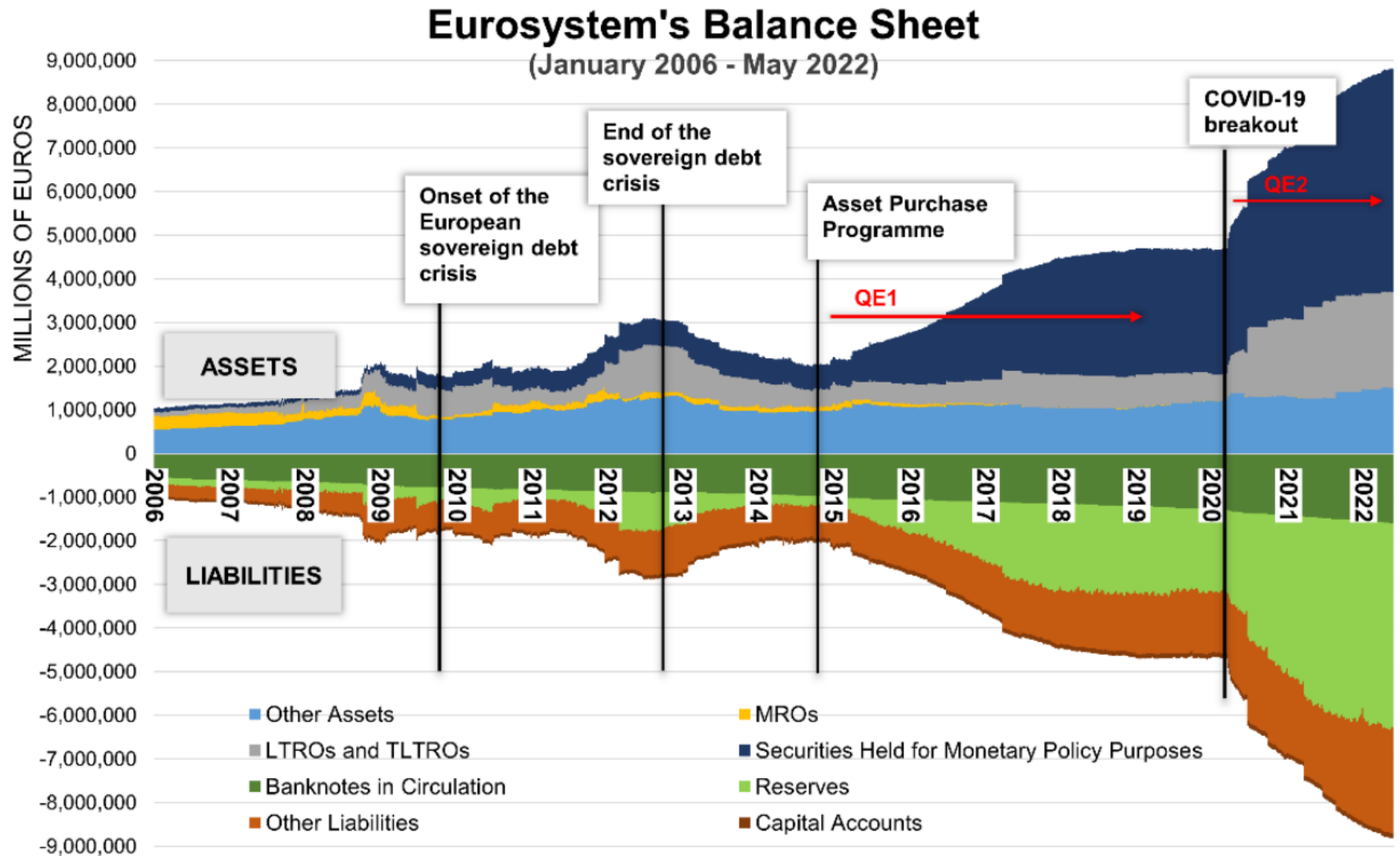


Decrease Interest on Reserves

- In Europe (ECB, Sweden and Denmark) enacted *negative* rates on deposits (btw 2012-2016)
- Idea: banks must lend to firms excess liquidity from QE rather than use deposit facility on Central bank
- Doubts wheter it was effective in incentivising banks to lend out to firms
- Banks in surplus (with higher ER) lower their profitability more, *less* (**NOT more**) lending



ECB Balance sheet



Leverage effect

A company produces with no debt initially and then with debt

$t=0$

$t=1$

Case 1: $i_{(I)} > i$: as $Lev \uparrow \Rightarrow ROE \uparrow$

Surplus = \$ 100
Equity = \$1,000
Debt = \$ 0
 $i = 9\%$
 $ROE = 10\% [= \$100 / \$1,000]$

Surplus = \$ 200
Equity = \$1,000
Debt = \$ 1,000
 $i = 9\%$
 $ROE = 11\% [=(\$200 - \$90) / \$1,000]$

Case 2: $i_{(I)} < i$: as $Lev \uparrow \Rightarrow ROE \downarrow$

Surplus = \$ 100
Equity = \$1,000
Debt = \$ 0
 $i = 11\%$
 $ROE = 10\% [= \$100 / \$1,000]$

Surplus = \$ 200
Equity = \$1,000
Debt = \$ 1,000
 $i = 11\%$
 $ROE = 9\% [=(\$200 - \$110) / \$1,000]$

